# Customer Behaviour Analysis

## About the Dataset

- **Type:** Retail / E-commerce Dataset
- **Nature:** Structured (Tabular)
- **Level:** Customer Transaction Level
- **Data Mix:**
  - Numerical → Age, Purchase Amount, Previous Purchases
  - Categorical → Category, Payment Method, Season

This dataset is used for **customer behavior analysis + business decision-making**

## Domain Background

In the **e-commerce industry**, companies track every customer interaction:

- What customers buy
- How often do they buy
- How much do they spend
- Whether they respond to discounts

## Why This Data Is Important

This data helps businesses:

- Identify **high-value customers**
- Understand **purchase patterns**
- Optimize **discount strategies**
- Improve **marketing campaigns**
- Increase **customer lifetime value (CLV)**

# Business Problem

How can a retail company understand customer purchasing behavior to increase revenue, improve customer retention, and optimize marketing strategies?

## Breaking Down the Problem

### Low Customer Retention

- Customers are not purchasing frequently

Need to understand:

- Who are frequent buyers?
- Who are one-time buyers?

### Ineffective Marketing Campaigns

- Discounts and promo codes may not be effective

Questions:

- Do discounts increase purchases?
- Which customers respond to promo codes?

### Product Demand Understanding

- Which category performs best?
- Which products generate the highest revenue?

### Payment Behavior Insights

- Which payment methods are preferred?
- Are digital payments increasing?

### Seasonal Demand Variation

- Does season impact sales?
- Which season drives maximum purchases?

### Customer Segmentation Problem

Need to identify:

- High-value customers
- Frequent buyers
- Discount-sensitive customers

# Data Dictionary

| Column Name | Description |
|---|---|
| Customer ID | Unique identifier for each customer |
| Age | Age of the customer |
| Gender | Gender of the customer (Male/Female/Other) |
| Item Purchased | Specific product purchased |
| Category | Product category (Clothing, Electronics, etc.) |
| Purchase Amount | Amount spent by the customer |
| Location | Customer’s geographic location |
| Size | Product size (S, M, L, etc.) |
| Color | Product color |
| Season | Season of purchase (Winter, Summer, etc.) |
| Review Rating | Customer rating for the product |
| Subscription Status | Whether customer is subscribed (Yes/No), Subscription Program |
| Shipping Type | Type of shipping selected |
| Discount Applied | Whether discount was applied |
| Previous Purchases | Number of past purchases |
| Payment Method | Mode of payment |
| Frequency of Purchases | Purchase frequency (Weekly, Monthly, etc.) |

# EDA- ( Exploratory Data Analysis using Python)

## Data Loading

What you did:

- Imported **pandas**
- Loaded dataset using:

```
#importing the dataframe/dataset
df=pd.read_csv('customer_shopping_behavior.csv')
```

Why did you do it:

- To bring raw data into a structured DataFrame
- Enables further analysis and transformations

```
df.head()
```

| | Customer ID | Age | Gender | Item Purchased | Category | Purchase Amount (USD) | Location | Size | Color | Season | Review Rating | Subscription Status | Shipping Type | Discount Applied | Promo Code Used | Previous Purchases | Payment Method |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | 2701 | 22 | Female | T-shirt | Clothing | 68.0 | California | XL | Olive | Winter | 3.2 | No | Standard | No | No | 36.0 | Cash |
| 1 | 521 | 51 | Male | Sunglasses | Accessories | 84.0 | South Carolina | M | White | Spring | 3.9 | Yes | Free Shipping | Yes | Yes | 20.0 | Debit Card |
| 2 | 3157 | 18 | Female | Shirt | Clothing | 50.0 | Montana | M | Black | Winter | 3.1 | No | 2-Day Shipping | No | No | 18.0 | Cash |
| 3 | 1687 | 22 | Male | Gloves | Accessories | 75.0 | Illinois | L | Red | Fall | 4.2 | No | Store Pickup | No | No | 25.0 | Cash |
| 4 | 2929 | 40 | Female | Jewelry | Accessories | 80.0 | Alabama | L | Yellow | Spring | 3.6 | No | Store Pickup | No | No | 17.0 | Credit Card |

## Initial Data Exploration

What you did:

- Viewed data using:
  - df.head()
  - df.tail()
- Checked dataset summary:
  - df.describe(include='all')
  - df.info()
- Checked missing values:
  - df.isnull().sum()

Why did you do it?

- To understand:
    - Data structure
    - Data types
    - Missing values
- This step helps identify **data quality issues early**

# Data Consistency Check (Category Correction)

## What you did:

- Checked unique values in:
    - Item Purchased
    - Category
- Found inconsistencies between **Item Purchased and Category**
- Created a **mapping dictionary**
- Updated category using:

## Why did you do it?

- Categories were inconsistent or incorrect
- Ensures:
    - Accurate grouping
    - Correct business insights

This is a **critical real-world data cleaning step**

# Handling Missing Values (Size Column)

## What you did:

Handled missing values based on category:

- Electronics → "Not Applicable."
- Accessories → "Not Applicable."
- Footwear → "Free Size."
- Clothing → Filled with **mode**

## Why did you do it?

- Different categories require **different business logic**
- Example:
    - Electronics don't have size → Not Applicable
    - Clothing → size is important → use mode

This shows **domain understanding**

# Handling Missing Values (Review Rating)

What you did:

- Checked:
  - Mean, median, mode
- Filled missing values using:
- Rounded values

Why did you do it:

- Instead of the global average, the **product-level mean is used**
- This keeps ratings **realistic and meaningful**

This is an **advanced imputation technique**

# Handling Missing Values (Purchase Amount)

What you did: Replaced with the mean value.

Why did you do it?

- Maintains **product-level pricing consistency**
- Avoids distortion in revenue analysis

# Handling Missing Values (Previous Purchases)

What you did: Replaced with the 0 (zero) value.

Why did you do it:

- Missing values imply **no previous purchase**
- Important for:
  - Customer segmentation
  - Loyalty analysis

# Duplicate Handling

What you did:

- Checked duplicates:
- Identified duplicates using Customer ID
- Removed duplicates:

Why did you do it?

- Ensures:
  - Each customer appears only once
  - No double-counting in analysis

Critical for **accurate reporting**

## Column Standardization

What you did:

- Converted column names:

df.columns = df.columns.str.replace(" ","_")
df.columns = df.columns.str.lower()

- Renamed column:

### Renaming the Headers

```
df.columns=df.columns.str.replace(" ","_")
df.columns=df.columns.str.lower()
```

Why did you do it?

- Makes column names:
  - Clean
  - SQL-friendly
  - Easy to use in Power BI

## Data Export to SQL Server

What you did:

- Installed required libraries:

pip install pyodbc sqlalchemy

- Connected to SQL Server
- Exported data:

Why did you do it?

- To move cleaned data into **SQL for further analysis**

- Enables:
  - Query-based insights
  - Integration with BI tools.

## Data Analysis using SQL Server

### 1. Which category generates the highest revenue?

| | category | total_revenue |
|---|---|---|
| 1 | Electronics | 486366.96 |
| 2 | Accessories | 194866.87 |
| 3 | Clothing | 183188.39 |
| 4 | Footwear | 111236.2 |
| 5 | Outerwear | 18524 |

### 2. Are discounts actually increasing purchase value?

| discount_applied | total_revenue | Avg_Purchase |
|---|---|---|
| No | 502241.64 | 182.24 |
| Yes | 491940.78 | 219.22 |

### 3. What is the total revenue generated by male and female customers?

| gender | total_revenue |
|---|---|
| Male | 390964.526238359 |
| Female | 352060.293140485 |
| Other | 251157.60467728 |

4. **Which customer used a discount but still spent more than the average purchase amount?**

| | customer_id | purchase_amount | discount_applied |
|---|---|---|---|
| 1 | 4568 | 276.343789954338 | Yes |
| 2 | 4256 | 734.47734939759 | Yes |
| 3 | 4289 | 1282.44 | Yes |
| 4 | 4576 | 831.344565217391 | Yes |
| 5 | 4319 | 685.231470588235 | Yes |
| 6 | 4708 | 735.773625 | Yes |
| 7 | 4584 | 685.231470588235 | Yes |
| 8 | 4607 | 557.24 | Yes |
| 9 | 4851 | 685.231470588235 | Yes |
| 10 | 4845 | 692.11 | Yes |

5. **Which are the top/bottom 5 products with the highest average review rating?**

| | item_purchased | avg_review_rating |
|---|---|---|
| 1 | Gloves | 3.86 |
| 2 | Sandals | 3.84 |
| 3 | Boots | 3.82 |
| 4 | Hat | 3.8 |
| 5 | Skirt | 3.79 |

| | item_purchased | avg_review_rating |
|---|---|---|
| 1 | Phone | 2.94 |
| 2 | Headphones | 3.08 |
| 3 | Watch | 3.1 |
| 4 | Bag | 3.17 |
| 5 | Laptop | 3.19 |

6. **Average purchase: Standard vs Express shipping**

| shipping_type | total_revenue | total_orders |
|---|---|---|
| Express | 358.12 | 1199 |
| Standard | 340.57 | 1201 |

7. **Do subscribed customers spend more? Compare average speed and total revenue between subscribers and non-subscribers.**

| subscription_status | total_customers | total_revenue | avg_spend |
|---|---|---|---|
| Yes | 1579 | 427245.4 | 270.58 |
| No | 3421 | 566937.02 | 165.72 |

8. **Top 5 products with highest discount usage %**

| item_purchased | total_number_of_times_sold | Number_of_times_sold_when_disocunt_applied | discount_percent |
|---|---|---|---|
| Laptop | 143 | 74 | 51.748251748251 |
| Phone | 171 | 88 | 51.461988304093 |
| Headphones | 166 | 85 | 51.204819277108 |
| Hat | 154 | 77 | 50.000000000000 |
| Watch | 157 | 78 | 49.681528662420 |

9. **Segment customers into new, returning, and loyal based on their total number of previous purchases, and show the count of each segment.**

| customer_segment | Customer_Count |
|---|---|
| Loyal Customers | 3085 |
| New Customer | 561 |
| Returning Customer | 1354 |

10. **What are the top 3 most purchased products within each category ?**

| category | item_purchased | most_purchased | RNK |
|---|---|---|---|
| Accessories | Jewelry | 171 | 1 |
| Accessories | Bag | 164 | 2 |
| Accessories | Belt | 161 | 3 |
| Accessories | Sunglasses | 161 | 3 |
| Clothing | Shirt | 315 | 1 |
| Clothing | Blouse | 171 | 2 |
| Clothing | Pants | 171 | 2 |
| Electronics | Phone | 171 | 1 |
| Electronics | Headphones | 166 | 2 |
| Electronics | Watch | 157 | 3 |
| Footwear | Shoes | 303 | 1 |
| Footwear | Sandals | 160 | 2 |
| Footwear | Sneakers | 145 | 3 |
| Outerwear | Jacket | 163 | 1 |
| Outerwear | Coat | 161 | 2 |

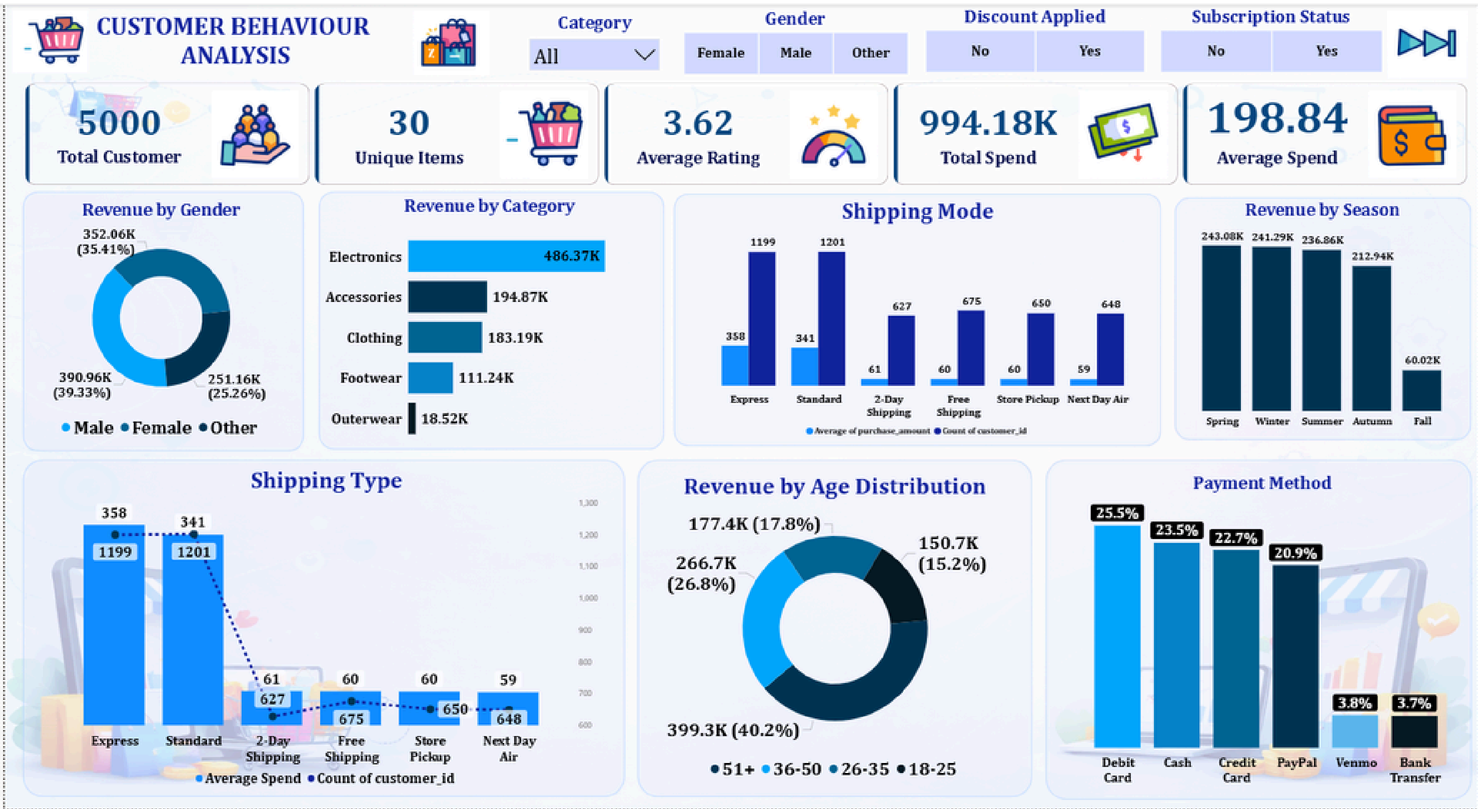
**11. Are customers who are repeat buyers (more than 5 previous purchases) also likely to subscribe?**

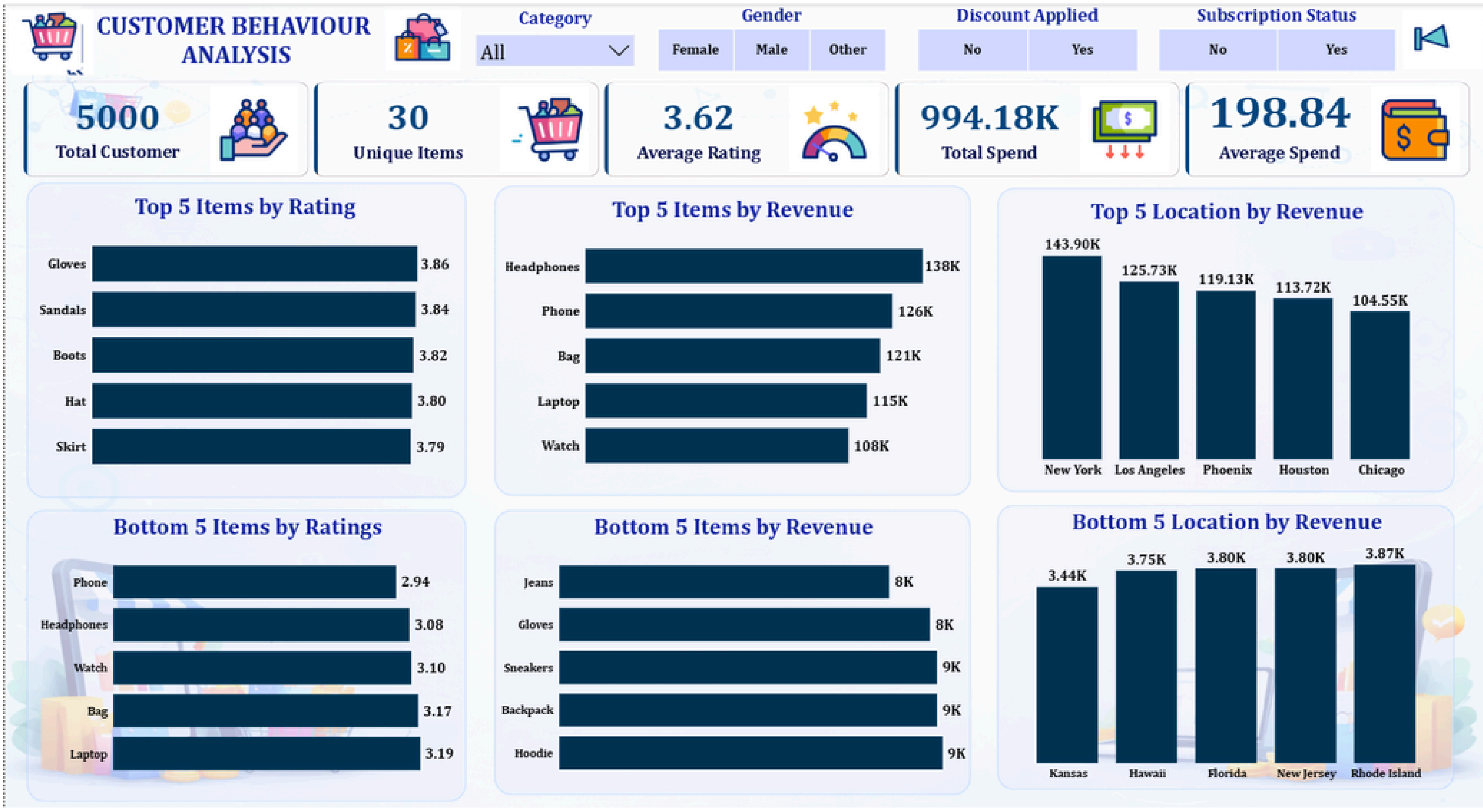
| Customer_type | subscription_status | Customer_count |
|---|---|---|
| Repeat Buyers | No | 2763 |
| Normal Buyers | Yes | 389 |
| Normal Buyers | No | 658 |
| Repeat Buyers | Yes | 1190 |

12. **What is the revenue contribution of each age group?**

| age_group | total_revenue |
|---|---|
| 51+ | 399345.32 |
| 35-50 | 266706.7 |
| 26-35 | 177421.93 |
| 18-25 | 150708.48 |

## Power BI Dashboard




CUSTOMER BEHAVIOUR ANALYSIS
Category
All
Gender
Female
Male
Other
Discount Applied
No
Yes
Subscription Status
No
Yes
5000
Total Customer
30
Unique Items
3.62
Average Rating
994.18K
Total Spend
198.84
Average Spend
Top 5 Items by Rating
Gloves 3.86
Sandals 3.84
Boots 3.82
Hat 3.80
Skirt 3.79
Top 5 Items by Revenue
Headphones 138K
Phone 126K
Bag 121K
Laptop 115K
Watch 108K
Top 5 Location by Revenue
New York 143.90K
Los Angeles 125.73K
Phoenix 119.13K
Houston 113.72K
Chicago 104.55K
Bottom 5 Items by Ratings
Phone 2.94
Headphones 3.08
Watch 3.10
Bag 3.17
Laptop 3.19
Bottom 5 Items by Revenue
Jeans 8K
Gloves 8K
Sneakers 9K
Backpack 9K
Hoodie 9K
Bottom 5 Location by Revenue
Kansas 3.44K
Hawaii 3.75K
Florida 3.80K
New Jersey 3.80K
Rhode Island 3.87K